

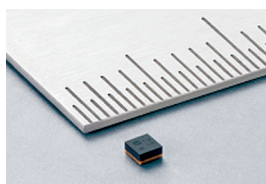
TRULY INNOVATIVE ELECTRONICS

INNOVATION UPDATES

Amongst numerous press releases of new products received by us, these are the ones we found worthy of the title *Truly Innovative Electronics*

High-frequency filter using XBAR technology

Murata Manufacturing Co., Ltd has begun mass production of the first high-frequency filter using XBAR technology, targeting wireless devices such



as smart-phones, wearables, laptops, and communication gateways.

Designed to support wireless systems such as 5G, Wi-Fi 6E, Wi-Fi 7, and 6G, the filter strikes a balance between performance and cost. It combines Murata's surface acoustic wave (SAW) filter work with XBAR technology from its subsidiary, Resonant Inc., to deliver low signal loss, high attenuation, and a wide bandwidth. The filter uses a bulk acoustic wave structure with comb-shaped electrodes and a piezoelectric thin film, enabling out-of-band attenuation and improved signal clarity at frequencies above 3GHz.

Murata Manufacturing Co., Ltd
<https://www.murata.com>

Jitter attenuating clocks

Skyworks Solutions has launched its latest timing products—the SKY63104/5/6 family of jitter attenuating clocks and the SKY62101 ultra-



low jitter clock generator, marking the industry's first devices capable of generating Ethernet and PCI Express (PCIe) spread spectrum clocks from a single chip. Built on Skyworks'

fifth-generation DSPLL and Multi-Synth technologies, these clocks offer flexible output with independently configurable settings for Ethernet, PCIe, and general-purpose clocking. Designed for applications in internet infrastructure (routers, switches, optical transport), data centers (Smart-NICs, AI accelerators, rack switches), and industrial systems (medical imaging, test tools, broadcast, and satellite communication), the devices support a wide frequency range and multiple output formats, allowing designers to simplify clock architectures and reduce component count across systems.

Skyworks Solutions
<https://www.skyworksinc.com>

Factory-integrated HD camera

ACEL Power has launched the first and only factory-integrated HD camera for electric outboard systems, now included in its Intelligent Electric Series. This camera provides real-time



visuals, diagnostics, and security data without

extra devices. It connects with ACEL's electric propulsion systems to enable remote monitoring, diagnostics, and operation. Users can access geotagged alerts and browser-based support through dual-mode Wi-Fi, GPS, BeiDou, and an optional 4G CAT 4 modem. The system features AI-based motion, intrusion, and face detection, along with privacy masking, all without relying on cloud servers. For images in glare, spray, or low light, the camera uses digital wide dynamic range (over 90dB), 2D/3D noise

reduction, and image processing.

ACEL Power
<https://acelpower.com>

CXL 3.0 memory controller

Primemas Inc. has announced customer samples of what it calls the first Compute Express Link (CXL) 3.0 memory controller. This controller is built using



the company's Hublet architecture. Unlike other CXL memory expansion controllers with fixed designs and memory limits,

Hublet uses a chiplet-based approach. It supports the CXL 3.0 specification, offering increased bandwidth and enhanced fabric capabilities compared to earlier versions. The SoC is made for workloads such as artificial intelligence, machine learning, and computing tasks that require steady memory access and data transfer. CXL 3.0 enables memory bandwidth and pooling, facilitating memory sharing and efficient system resource utilisation.

Primemas Inc.
<https://www.primemas.com>

Fan-on-a-chip for XR smart glasses

XMEMS Labs has expanded its silicon-based μ Cooling fan-on-a-chip platform to smart glasses used in extended reality (XR), becoming the first to integrate an active cooling system directly into the frame of wearable displays. The system provides local cooling without altering the shape or appearance of the glasses, thereby helping to maintain comfort and safety during prolonged use. In thermal tests at a 1.5W power